C3.4 Electrolysis

Summary

Decomposition of a liquid during the conduction of electricity is a chemical reaction called electrolysis. This section explores the electrolysis of various molten ionic liquids and aqueous ionic solutions.

Underlying knowledge and understanding

Learners should be familiar with ionic solutions and solids.

Common misconceptions

A common misconception is that ionic solutions conduct because of the movement of electrons. Another common misconception is that ionic solids do not conduct electricity because electrons cannot move.

Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers.
All other statements will be assessed in both Foundation and Higher Tier papers.

Reference	Mathematical learning outcomes	Mathematical skills
CM3.4i	arithmetic computation and ratio when determining empirical formulae, balancing equations	M1a, M1c

Topic content			Opportunities to cover:		Practical suggestions
Learning outcomes		To include	Maths	Working scientifically	
C3.4a	recall that metals (or hydrogen) are formed at the cathode and non-metals are formed at the anode in electrolysis using inert electrodes	the terms cations and anions		WS1.4a	
C3.4b	predict the products of electrolysis of binary ionic compounds in the molten state	compounds such as $NaCl$	M1a, M1c	WS1.2a, WS1.2b, WS1.2c, WS1.4a, WS2a, WS2b	
C3.4c	describe competing reactions in the electrolysis of aqueous solutions of ionic compounds in terms of the different species present	the electrolysis of aqueous $NaCl$ and $CuSO_4$ using inert electrodes	M1a, M1c	WS1.4a	Electrolysis of sodium chloride solution. (PAG C2) Electrolysis of copper sulfate solution. (PAG C2)
C3.4d	describe electrolysis in terms of the ions present and reactions at the electrodes	the equations and **half equations** of the reactions at the electrodes	M1a, M1c		
C3.4e	describe the technique of electrolysis using inert and non-inert electrodes				